

Enabling High-Throughput Routing for LEO Satellite Broadband Networks: A Flow-Centric Deep Reinforcement Learning Approach

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Abstract—Routing optimization within a low-Earth orbit (LEO) satellite broadband network (LSBN) has seen advancements through deep reinforcement learning (DRL) approaches in academia. Nonetheless, a crucial aspect often overlooked in these approaches pertains to the inference time of deep neural network (DNN) models during the routing of packets. Our investigation reveals that this oversight can significantly impair routing throughput in LSBN. In response, this article innovatively proposes a decentralized flow-centric DRL approach, shifting the focus from routing individual packets to entire traffic flows. To align with the large-scale feature of LSBN, we embrace a fully distributed architecture for flow-centric routing, which is modeled as a partially observable Markov decision process. In this construct, each satellite operates as an independent agent, locally classifying flows following a tailor-designed definition, and is responsible for forwarding a flow to an adjacent satellite based on its internal policy. Notably, the DNN inference is conducted only once on each agent to determine the route for the initial packet of a specific flow; subsequent packets are directed along the same route. Recognizing the potential impact of dynamic LSBN topologies on routing performance, we also introduce an adaptive flow routing update scheme. This scheme is completely free from LSBN environment modeling and aims to bolster the efficacy of the flow-centric approach. Comparative experiments showcase the superiority of the proposed approach over baseline algorithms across various metrics. Consequently, the flow-centric DRL approach can enable high-throughput traffic transmission for LSBN.

Index Terms—Deep reinforcement learning (DRL), flow-centric routing, high-throughput, inference time, low-Earth orbit (LEO) satellite network.

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I. INTRODUCTION

LARGE-SCALE low-Earth orbit (LEO) satellite networks, such as the *Starlink* constellation proposed by SpaceX [1], have recently successfully attracted massive research interests from both industry and academia. LEO satellite broadband networks (LSBNs) provide seamless coverage on Earth and are regarded as important supplements to terrestrial networks, playing essential roles in the forthcoming 6G communication systems [2], [3]. However, LSBN exhibits high-dynamic, large-scale and, high-throughput characteristics, rendering most conventional routing algorithms designed for legacy terrestrial networks not directly applicable in LSBN. First, the high mobility of LEO satellites causes time-varying topologies [4] and traffic distribution, which make static routing ineffective in LSBN [5]. Second, LSBN covers a vast spatial scale and typically consists of a substantial number of satellites [6], leading to heightened communication overhead and notable response delays [7], [8], [9]. Third, LEO satellites possess remarkably high throughput in data forwarding [10], which in turn imposes elevated demands on routing decision-making speed. Therefore, it is very essential to specifically design effective routing algorithms tailored for LSBNs.

In recent years, deep reinforcement learning (DRL)-based artificial intelligence technologies are increasingly used to optimize routing problems across various scenarios, encompassing Internet of Things (IoT) [11], [12], [13], [14], SDN [15], [16], [17], [18], [19], [20], [21], mobile edge computing [22], [23], and more. DRL utilizes deep neural networks (DNNs) to approximate policy or value functions, thereby adapting to dynamic and complex environments through learning and self-adjustment [24], [25]. Relevant studies mainly fall in to two categories: 1) *packet routing* [26], [27], [28], [29], [30], [31], [32] and 2) *flow routing* [15], [16], [17], [18], [19], [20], [21]. DRL-based packet routing approaches are typically implemented in distributed manners, which exhibit scalability and flexibility advantages in large-scale scenarios. Consequently, proposals employing DRL to tackle routing issues in LSBN primarily center around distributed packet routing [31], [32]. Xu et al. [31] presented a fully distributed routing algorithm with spatial location assistance. Building upon this, an innovative routing approach known as

FDR-MARL is introduced, enabling each satellite to determine the optimal routing decision for each received packet. In addition, Liu et al. [32] proposed a DRL-based energy-efficient routing protocol (DRL-ER). Similar to FDR-MARL, DRL-ER is designed to select next hops for all packets via the inferences of DNN on each satellite. The preliminary evaluation results show that DRL can outperform legacy routing algorithms in LSBNs.

Despite prior research efforts, applying DRL in LSBN still faces a significant issue. These studies simply assume that the routing decision for a packet can be made immediately upon a router receiving it. However, given the constrained energy and computation resources on LEO satellites, it is impractical to entirely dismiss the consideration of DNN inference time [33]. Furthermore, LEO satellites are continually advancing toward extremely high throughput via laser links [10], [34], far exceeding the capacity of DRL-based routers. When the DNN inference speed lags behind the packet forwarding rate onboard of a satellite, it will increase packet transmission delay and packet loss ratio, ultimately limiting throughput in LSBN. Thus, ignoring the DNN inference time may threaten the correctness of conclusions drawn in prior works.

Given the limitation, we pivot our focus to DRL-based flow routing, which involves conducting DNN inferences on traffic flows rather than individual packets, thus mitigating the adverse effects of DNN inference time. Most flow routing approaches are designed using centralized architectures, with SDN being widely adopted [18]. The DRL agent is typically deployed in the central controller to acquire global views and decide full-path routes for flows, with minimal instances of forwarding devices independently routing flows. However, due to the distinctive characteristics of LSBN, utilizing DRL for flow routing still encounters the following challenges.

- 1) The large-scale nature of LSBN leads to heightened communication overhead and response delays for central control [7], [8], undermining the adaptivity and scalability of existing DRL-based flow routing approaches.
- 2) LSBN caters to a worldwide user base and may establish millions of flows in a short time, introducing substantial computational overhead for flow routing.
- 3) The dynamism in LSBN weakens the sustainability of established flow tables. It is difficult for satellites to independently determine when to update routes by modeling the complex dynamic environments [5].

To address the throughput limitation posed by DNN inference time, we propose a decentralized flow-centric DRL approach that aligns with the distinctive characteristics of LSBN. This represents an early attempt to propose DRL within a fully distributed architecture for routing traffic flows. The flow-centric routing is formulated as a partially observable Markov decision process (POMDP)-based multiagent decision problem, where each LEO satellite acts as an independent agent responsible for forwarding flows to neighboring satellites based on its own policy. To prevent the explosion in the amount of established flows, the flow definition is tailored in a distributed manner, where each satellite locally classifies flows based on the destination satellite information of received packets. It is noted that the DNN inference is conducted only once

to decide the initial packet routing for a specific flow on each agent, after which all subsequent packets of this flow will be directly forwarded following the established route. Moreover, given the high-dynamic nature of LSBN, which potentially impacts routing performance, we introduce an adaptive and model-free routing updating scheme to enhance the proposed flow-centric approach. In summary, the main contributions are listed as follows: 1) we provide pioneering insights into the impact of DNN inference time on DRL-based packet routing performance, derived from a preliminary experiment and quantitative analysis; 2) we propose an innovative decentralized flow-centric DRL approach with a tailor-designed flow definition, enabling each satellite to independently conduct DNN inferences for routing local-defined flows as opposed to individual packets; and 3) we further derive an adaptive flow routing updating scheme, which is entirely free from the complex network models and relies solely on perceiving delay jitters, to enhance our proposed flow-centric DRL approach.

The remaining part of this article is organized as follows. Section II investigates the related work. Section III describes the problem by experimentally illustrating the negative impact of considering DNN model inference time. In Section IV, the system model and the relevant assumptions are presented. In Section V, a flow-centric DRL approach is elaborated to conduct routing for flows. Section VI further derives an adaptive flow routing updating scheme. Comparative simulation experiments are carried out to evaluate the performance of the proposed flow-centric DRL approach in Section VII. Finally, Section VIII concludes this article and gives the outlook.

II. RELATED WORK

A. DRL-Based Routing Approaches

The application of DRL to tackle routing challenges is initially introduced by Boyan and Littman [35], employing Q -learning to assist in selecting of the next hop node for each packet. Subsequently, plenty of approaches have been proposed by utilizing DRL to optimize traffic routing while considering the evolving nature of networks and user requirements, which can be broadly classified into the following two categories:

1) *DRL-Based Packet Routing*: These approaches are typically implemented in a distributed manner, particularly in large-scale systems. To address the overwhelming dynamics of underwater environments, Di Valerio et al. [26] proposed a DRL-based new packet routing scheme known as CARMA, enabling rapid adaptation of relay selection to the changing conditions of underwater channels. The packet routing in aeronautical ad-hoc networks (AANETs) is challenging due to their high-dynamic topology. To minimize end-to-end packet delay in AANETs, Liu et al. [27] developed a deep Q -network (DQN), which could capture the relationship between the optimal routing decision and the local geographic information. You et al. [28] integrated a long short-term memory (LSTM) into multiagent DRL to address the curse of dimensionality. It can provide a more comprehensive representation of dynamic network states for the fully distributed routing. In addition, considering the graph nature of network topology,

Mai et al. [29] designed a multiagent DRL framework in combination with a graph attention network (GAT), where routers learn from interactions with the network to make better routing decisions for the future. He et al. [30] pointed out that centralized DRL could mismatch the real-time property of routing, and they proposed a real-time and distributed learning approach, i.e., RTHop, to realize a hop-by-hop routing approach.

2) *DRL-Based Flow Routing*: For this kind of approaches, DRL agents are generally deployed in centralized SDN controllers to obtain a global view of the network. Sun et al. [15] introduced ScaleDeep, a scalable DRL-based flow routing scheme for SDN. Leveraging partial control over critical nodes and DRL techniques, ScaleDeep significantly improves routing performance and resilience to topology changes. In addressing the challenge of optimizing forwarding paths for traffic flows with multiple QoS requirements in real-time, Liu et al. [16] proposed DRL-OR-S, a highly scalable online routing algorithm using a novel graph-based actor-critic network architecture. Additionally, Chen et al. [17] developed an DRL-based routing algorithm known as RL-Routing to address SDN traffic engineering concerns related to throughput and transmission delay, eschewing the complexities of network modeling in favor of experiential learning. Considering the requirements of efficient routing protocols with low latency and high security in SDN-IoT, Guo et al. [19] proposed a DRL-based QoS-aware secure routing protocol (DQSP). DQSP can extract knowledge from history traffic demands and dynamically optimize the routing policy. Moreover, Casas-Velasco et al. [20] defined a DRL-based routing algorithm in SDN named DRSIR, which considers path-state metrics to produce proactive, efficient, and intelligent routing that adapts to dynamic traffic changes.

B. Routing in LEO Satellite Networks

Numerous studies have explored routing problems in LEO satellite networks using conventional approaches. Werner [36] proposed a DT-DVTR routing scheme for LEO satellite networks, employing an asynchronous transmission mode (ATM). This scheme models network connections using a space-time graph and identifies the least costly paths through a shortest path algorithm. Ekici et al. [37] introduced a datagram routing algorithm for LEO satellite networks, which focuses on generating paths with minimum propagation delay. To achieve a more balanced traffic distribution across all links, Taleb et al. [38] proposed an explicit load balancing (ELB) scheme that takes into account the traffic load of each satellite. If a satellite's load becomes excessively high, it notifies its neighboring satellites and requests them to reduce their transmission rates. To dynamically adjust the routing strategy in N GEO satellite IP networks, a traffic-light-based intelligent routing strategy (TLR) was proposed [39]. TLR utilizes multiple traffic lights to present the status of nodes and integrates preconfigured routing with dynamic adjustments. In addition, Li et al. [40] introduced a state-aware and load-balanced (SALB) model, which addresses uneven traffic distribution and congestion by estimating link states, adjusting queuing delay weights, and using an efficient shortest path

tree algorithm for routing updates. In [41], a branch-and-price algorithm was designed to efficiently minimize the intersatellite resource consumption.

Several studies investigated the innovative software defined satellite networks (SDSNs) [42]. Li et al. [43] proposed an SDSN architecture that combines the information centric networking (ICN) and SDN for satellite-terrestrial integrated networks, aiming to enhance the efficiency of satellite routing. Hu et al. [44] made use of SDN features and leveraged unicast and multicast traffic engineering methods to enhance the intersatellite link (ISL) performance in LSBNs. While these explorations offer insights, further development and validation are continually required in large-scale scenarios.

With the rapid expansion of LSBNs and the success of DRL applications in terrestrial networks, researchers have increasingly focused on harnessing DRL to enhance routing performance in LSBNs. Wang et al. [45] introduced the GRouting algorithm, combining graph neural networks (GNNs) and DRL to dynamically make optimal routing decisions for each data packet in LSBNs. Zuo et al. [46], [47] introduced intelligent decentralized and load balancing routing algorithms, both of which dynamically select the optimal next-hop node based solely on the conditions of neighboring satellites. To minimize the average delivery time of packets in LSBNs, Xu et al. [31], [48] formulated the routing problem to be a POMDP, and developed a distributed dynamic packet routing algorithm based on multiagent DRL. In addition, Liu et al. [32] presented the DRL-ER protocol, a novel DRL-based energy-efficient routing protocol. It solves the battery energy imbalance problem in LEO satellite networks, with keeping the packet transmission delay below an acceptable upbound. In response to the scale and dynamic topology features of LSBNs, Huang et al. [49] proposed a Q -learning-based dynamic distributed routing scheme, which adopts a multiobjective optimization method to find the optimal next hop for forwarding data packets. In a different approach, a centralized routing manager is introduced for medium Earth orbit (MEO) satellites in [50], presenting the EEDRL-Dijkstra routing algorithm. This algorithm is designed to address stochastic routing request events and achieve long-term optimization for LEO satellite networks in 6G.

C. Research Gaps

In the literature, existing research on DRL-based packet routing approaches does not explore the influence of DNN inference time on routing performance. On the one hand, DRL is widely recognized for real-time decisions, which effectively match the packet forwarding speed in common wireless networks, such as underwater networks [26] and AANETs [27]. DNN inference time is not a significant concern in these studies. However, on the other hand, LSBN uses laser ISLs, providing throughput capabilities far surpassing those of other wireless networks. It is crucial to note that the throughput of satellites may face constraints imposed by the lagging DNN inference time. Hence, in prior LSBN studies [31], [32], [45], [46], [47], [48], [49], the oversight of inference time will undermine the feasibility of DRL-based routing in high-throughput LSBNs.

Moreover, DRL-based flow routing approaches [15], [16], [17], [19], [20] provide insights toward mitigating the impact of inference time. Most of them are designed with a reliance on centralized architectures, especially SDN, to alleviate complexity and ensure consistency. However, emerging LSBN possesses large-scale characteristics far exceeding those of common networks. This introduces unsolved challenges for centralized DRL implementations, including increased communication overhead and limited scalability. In practice, the development of flow routing protocols does not necessarily mandate central controllers. However, there is a noticeable research gap in exploring DRL within fully distributed architectures for flow routing in large-scale networking scenarios.

III. MOTIVATION

When applying DRL to address routing problems in LSBN, a satellite serving routing functions is typically considered as an agent, while its surroundings constitute the environment. The action of each satellite involves selecting one of its neighboring satellites to forward received packets. To combat the challenge of dimension explosion within the state space, DRL employs DNN models to approximate value tables and replace policy functions, significantly streamlining the routing decision-making process.

The architectural design of DRL-based routers is different from that of conventional routers. The conventional router manages two tables, a routing table and a forwarding table. The routing table undergoes long-term updates based on routing protocols, while the forwarding table, derived from the routing tables, is typically deployed at network interface cards to achieve fast packet forwarding beyond Gbps. However, the majority of research [26], [27], [28], [29], [30], [31], [32] on DRL-based routers do not adopt the above two-table architecture. Instead, they utilize DNN inferences to make routing decisions on a per-packet basis according to current network states, thereafter forwarding the packets accordingly. These studies have demonstrated that DRL-based packet routing approaches can substantially enhance routing performance. However, most of these approaches have naively assumed that a routing decision can be made instantaneously upon packet reception, neglecting the practical time taken for DNN model inference, which can adversely affect routing performance in LSBN.

To quantitatively analyze the impact on LSBNs, we investigate a typical DRL-based packet router. We assume that the router, deployed on an LEO satellite as a DRL agent, incorporates a DNN model for making routing decisions. The DNN model takes an $\mathcal{N}_s$ -dimensional state vector as input and produces an output vector containing $\mathcal{N}_a$ routing decisions. The DNN model consists of $\mathcal{L}_{nn}$ fully connected hidden layers, with each layer containing $\mathcal{H}_l$ neurons, where $l \in \{0, 1, \dots, \mathcal{L}_{nn} - 1\}$. Thus, the time complexity of model inference can be expressed as

$$\mathcal{O}\left(\mathcal{N}_s \mathcal{H}_0 + \mathcal{N}_a \mathcal{H}_{\mathcal{L}_{nn}-1} + \sum_{l=1}^{\mathcal{L}_{nn}-1} \mathcal{H}_{l-1} \mathcal{H}_l\right). \quad (1)$$

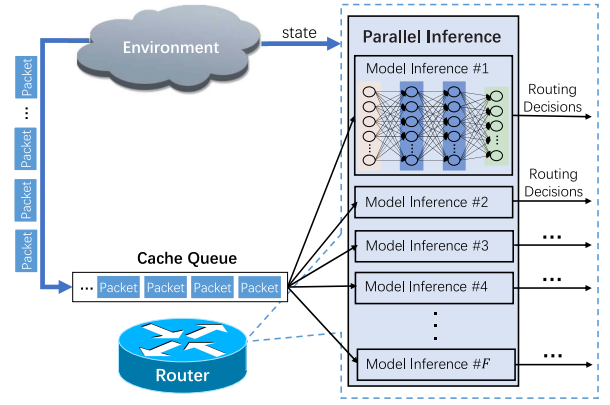


Fig. 1. DRL-based packet routing decision making process in a router.

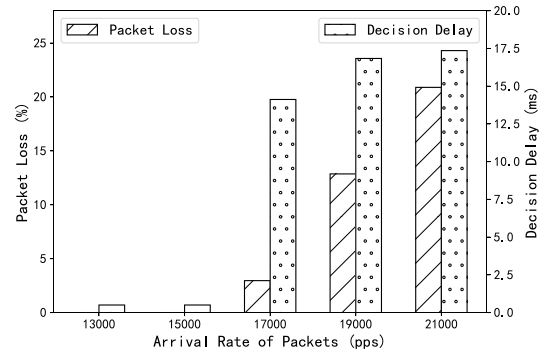


Fig. 2. Experimental results against different packet arrival rates.

For the sake of generality, we assume that the router is equipped with a multicore processor capable of performing up to F model inferences in parallel. As depicted in Fig. 1, when the router receives a packet, it performs DNN model inference to make routing decisions. If the number of parallel inferences reaches the upper bound F , the packet is queued in cache, awaiting processor resources to become available. If the cache is running out, subsequent packets will be dropped, resulting in an increased packet loss ratio.

Experiments are conducted on a high-performance computer to simulate the DRL-based packet routing process on the router, incorporating parallel inferences and packet caching. The parameter values for the DNN model are given below. The values 3 and 256 are assigned to $\mathcal{L}_{nn}$ and $\mathcal{H}_l$, respectively. We set $\mathcal{N}_s$ to 12 and $\mathcal{N}_a$ to 4, with F assumed to be 8. The maximum capacity of the cache M is set at 300 packets, and the packet arrival rate ranges from 13 000 to 21 000 packets per second (pps). The evaluation results are illustrated in Fig. 2. As the packet arrival rate increases, the router struggles to meet the real-time demands of DNN model inference, leading to significant packet routing decision delays and an unacceptable packet loss ratio. For instance, when the packet arrival rate reaches 21 000 pps (approximately 300 Mb/s, assuming a packet size of 1500 bytes), the packet routing decision delay and packet loss ratio increase to 17.5 milliseconds (ms) and 21%, respectively.

With the advancement of laser communication technology, ISLs in typical LSBNs are evolving toward greater network bandwidth and enhanced throughput capabilities. Laser ISLs have achieved speeds of 5.6 Gb/s over distances up to 4900 km between LEO satellites [51]. Companies, such as Mynaric [52] and Tesat [53], are marketing multi-Gbps ISL solutions for LSBNs, with intentions to provide 100+ Gb/s offerings in the near future. Thus, the conjecture [54] concerning ISL capacities of tens of Gbps is academically sound. Unfortunately, considering the results of the aforementioned experiment, a significant issue is highlighted: the DNN model inference speed seriously degrades the routing performance of LEO satellites and falls far short of meeting the high-throughput demands in LSBNs.

Researchers have recently proposed diverse methods to shorten DNN model inference time, such as weight and branch pruning, tensor quantization, and knowledge distillation. Nevertheless, these methods usually lead to compromised model accuracy [55] and inconspicuous effects. Besides, some massive accelerators, such as GPUs [56] and ASIC [57], are utilized to enhance computation resources. We also observe the promising neural processing unit (NPU) specifically designed for accelerating DNN inference on embedded devices [58]. But the feasibility of practically deploying these more powerful accelerators onboard of LEO satellites is still under investigation. In addition to enhancing hardware capabilities, there is still ample opportunity for improving routing approaches. The two aspects are not mutually exclusive. Therefore, the purpose of this research is to elaborate a more efficient DRL approach for high-throughput traffic routing in LSBNs, which can significantly eliminate the negative impact caused by the cumulative DNN model inference delay.

IV. SYSTEM MODEL

A. Network Model

In this study, we delve into the realm of a large-scale LSBN, comprising N_{orb} orbital planes. Within each plane, a set of N_{sat} evenly distributed LEO satellites traverse at high velocities along their designated orbital trajectories. The inherent design of this network enables the provision of wireless broadband access services seamlessly across the globe to a multitude of ground users. The ensemble of LEO satellites within a LSBN is represented as $\mathbb{S} = \{\text{Sat}_1, \text{Sat}_2, \dots, \text{Sat}_{\text{total}}\}$, where *total* signifies the complete count of LEO satellites. Drawing inspiration from established LEO satellite constellations, e.g., Iridium [59], this article assumes a *Walker Star* constellation configured LSBN. More preciously, we assume a grid-style configuration for the ISLs. This configuration permits each LEO satellite to establish four distinct bi-directional ISLs with its neighboring satellites.

Specifically, the ISL arrangement consists of two intraplane links connecting the upper and lower satellites within the same orbital plane, and two interplane links connecting satellites situated on adjacent orbits, to the left and right, respectively. The notation $\text{link}_{i,j}$ is employed to symbolize the link originating from Sat_i and terminating at Sat_j . This particular interconnection pattern is termed as “+Grid,” a nomenclature

reflective of its underlying geometric structure. It has gained widespread acceptance as a standard both within the industry and in relevant academic literature [34], [60].

Unlike the geosynchronous Earth orbit (GEO), LEO introduces a heightened level of dynamism to satellite constellations [54]. This dynamism arises from the rapid relative motion observed among LEO satellites situated in adjacent or closely positioned orbital planes. As a result, ISLs that establish connections between LEO satellites in these neighboring orbital planes may undergo frequent toggling. In contrast, satellites within the same orbital plane exhibit minimal relative velocities, thus ensuring the stability of ISLs within that intraplane configuration.

B. User Model

The user model in our study encompasses two critical aspects: 1) the *user distribution model*, which delineates the geographic dispersion of users on Earth, thereby shaping the network traffic distribution, and 2) the *user behavior model*, which elucidates how users generate traffic packets.

1) *User Distribution Model*: LSBNs offer global coverage without being affected by geographical constraints. Consequently, any terminal node located on the ground or in the sky is a potential user of LSBN. However, user distribution across the Earth is not uniform. By referring to the current world population data set released by NASA [61], we partition the Earth surface into r contiguous regions and denote this set of regions as $\mathbb{R} = \{R_1, R_2, \dots, R_r\}$, each characterized by a unique user population density. Users in a given region, for instance R_m where $m \in \{1, 2, \dots, r\}$, are randomly distributed. The probability density function (PDF) governing this distribution is expressed as

$$f(x, y; R_r) = \begin{cases} \frac{1}{\text{Square}(R_r)}, & (x, y) \in R_r \\ 0, & \text{otherwise.} \end{cases} \quad (2)$$

Here, $f(x, y; R_r)$ represents the probability of a user being located at (x, y) within the region R_r , and *Square*($\cdot$) indicates the square of regions.

2) *User Behavior Model*: Network traffic originates from user behavior. To create a more accurate traffic model for LSBN, we assume that users independently generate packets at regular intervals. The time durations between two consecutive packets sent by a user adhere to a negative exponential distribution with the following PDF:

$$f(x; \lambda) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

where λ is the expected number of request occurrences by a single user per unit of time. The packets sent by users are subsequently routed to their intended destination nodes through LEO satellite routes within LSBN. It is noted that user mobility is significantly lower in comparison to satellite movement and is therefore considered negligible in this study.

C. Routing Model

The research employs a fully distributed network routing framework, as depicted in Fig. 3. By considering LEO#23, the routing framework is segmented into two distinct processes.

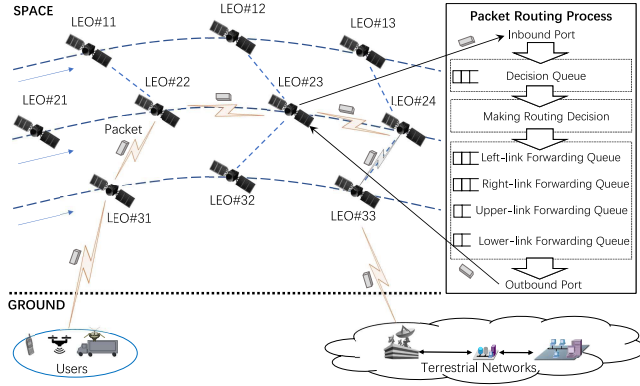


Fig. 3. Fully distributed routing framework in LEO satellite networks.

1) *Decision Process*: Upon receiving a packet, each satellite undertakes a decision process to determine the subsequent-hop node based on specific routing protocols. In some cases, these routing strategies may involve notable time complexities. Acknowledging limited computational capabilities, a decision queue is implemented, following the first-in-first-out (FIFO) principle. In this queue, incoming packets are temporarily held, awaiting the availability of computation resources for the decision-making process.

2) *Forwarding Process*: Following the decision-making process, an LEO satellite proceeds to forward the data packet to the designated next-hop satellite via the relevant ISL. If the network's bandwidth resources approach full utilization, the data packet is placed into a FIFO forwarding queue. Here, it awaits the eventual release of the necessary bandwidth resources, enabling further transmission. The aforementioned procedure repeats for subsequent LEO satellites in the packet forwarding path if the next-hop satellite is not the intended destination node. It's noteworthy that both the decision queue and forwarding queue possess finite caching capacities. Upon reaching their threshold capacities, any ensuing packets are subject to being discarded. This routing model encapsulates the intricate dynamics of decision-making and packet forwarding within the LSBN, ensuring effective data transmission while taking into account the computational limitations and available bandwidth resources.

D. Delay Model

The delay model within this framework encompasses two integral components: 1) the *decision delay* and 2) the *forwarding delay*. The decision delay signifies the time elapsed from the packet's reception until the routing decision is finalized. Conversely, the forwarding delay pertains to the temporal interval between the moment of routing decision formulation and the subsequent-hop satellite's receipt of the packet. More specifically, for routing packet $\mathcal{P}_k$ on LEO satellite Sat_i , the decision delay $\mathcal{D}_{i,k}^{dec}$ consists of two components: 1) the *decision-queuing delay* $\mathcal{D}_{i,k}^{q,dec}$ and 2) the *decision-making delay* $\mathcal{D}_{i,k}^{m,dec}$. The forwarding delay to Sat_j is denoted as $\mathcal{D}_{i,j,k}^{fwd}$, inclusive of the *forwarding-queuing delay* $\mathcal{D}_{i,j,k}^{q,fwd}$, the *transmission delay* $\mathcal{D}_{i,j,k}^{trans}$, and the *propagation delay* $\mathcal{D}_{i,j,k}^{pro}$.

The calculation formulas for these delay components are elaborated below.

1) *Decision-Making Delay*: The decision-making delay is directly impacted by the complexity of the routing strategy employed and the computational capacity of the routers. When the strategy aligns with predefined routing tables or rules, the decision-making delay can be deemed negligible. However, when this is not the case, the decision-making delay $\mathcal{D}_{i,k}^{m,dec}$ is computed as follows:

$$\mathcal{D}_{i,k}^{m,dec} = \frac{C_k}{f_{i,k}}. \quad (4)$$

Here, C_k represents the count of CPU cycles necessary for routing decision-making for packet $\mathcal{P}_k$, and $f_{i,k}$ denotes the allocated computation capabilities (in CPU cycles/s) for $\mathcal{P}_k$ by LEO satellite Sat_i .

2) *Decision-Queuing Delay*: The decision-queuing delay adheres to the $M/D/C/N$ queuing model. In this model, M signifies that data packets arrive at LEO satellite Sat_i through a Poisson process with an expectation of λ_i^{dec} . D indicates a constant value for the decision time for each packet, set as $\mathcal{D}_{i,k}^{m,dec}$. C represents the capacity for satellite Sat_i to concurrently make routing decisions for up to C_i^{dec} packets. N denotes the maximum capacity of the decision queue, set as N_i^{dec} . Based on established findings regarding the $M/D/C/N$ queue model, the decision-queuing delay for packet $\mathcal{P}_k$ on LEO satellite Sat_i is approximated as follows:

$$\mathcal{D}_{i,k}^{q,dec} = \frac{\mathcal{D}_{i,k}^{m,dec}}{C_i^{dec} - \lambda_i^{dec} \mathcal{D}_{i,k}^{m,dec}} = Q_{i,k}^{dec} \frac{\mathcal{D}_{i,k}^{m,dec}}{C_i^{dec}} \quad (5)$$

where $Q_{i,k}^{dec}$ denotes the count of queuing packets preceding packet $\mathcal{P}_k$.

3) *Transmission Delay*: Given that the network bandwidth allocated to transmit packet $\mathcal{P}_k$ on $link_{i,j}$ is $B_{i,j,k}$, the transmission delay $\mathcal{D}_{i,j,k}^{trans}$ can be calculated by

$$\mathcal{D}_{i,j,k}^{trans} = \frac{S_k}{B_{i,j,k}} \quad (6)$$

where S_k indicates the size of packet $\mathcal{P}_k$.

4) *Forwarding-Queuing Delay*: If the bandwidth is temporarily unavailable on $link_{i,j}$, packet $\mathcal{P}_k$ will be put into a forwarding queue in the cache for $link_{i,j}$, introducing the forwarding queuing delay $\mathcal{D}_{i,j,k}^{q,fwd}$

$$\mathcal{D}_{i,j,k}^{q,fwd} = \frac{\mathcal{D}_{i,j,k}^{trans}}{C_{i,j}^{fwd} - \lambda_{i,j}^{fwd} \mathcal{D}_{i,j,k}^{trans}} = Q_{i,j,k}^{fwd} \frac{\mathcal{D}_{i,j,k}^{trans}}{C_{i,j}^{fwd}}. \quad (7)$$

Here, $\mathcal{D}_{i,j,k}^{trans}$ represents the transmission time on $link_{i,j}$, $\lambda_{i,j}^{fwd}$ denotes the expectation of packet arrival rate, $C_{i,j}^{fwd}$ is the capacity for concurrently forwarding packets, and $Q_{i,j,k}^{fwd}$ denotes the count of queuing packets preceding packet $\mathcal{P}_k$.

5) *Propagation Delay*: The spatial coordinates of satellites Sat_i and Sat_j at time t are denoted as (x_i, y_i, z_i) and (x_j, y_j, z_j) , respectively. Thus, their spatial distance $Dis_{i,j}$ can be calculated as

$$Dis_{i,j} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2 + (z_i - z_j)^2}. \quad (8)$$

By approximating the propagation distance of link_{*i,j*} with $Dis_{i,j}$, the propagation delay $\mathcal{D}_{i,j,k}^{\text{pro}}$ can be represented as

$$\mathcal{D}_{i,j,k}^{\text{pro}} = \frac{Dis_{i,j}}{c} \quad (9)$$

where c represents the speed of the light.

In summary, the delay for routing packet $\mathcal{P}_k$ on satellite Sat_i can be denoted as

$$\mathcal{D}_{i,k} = \underbrace{\mathcal{D}_{i,k}^{q,\text{dec}} + \mathcal{D}_{i,k}^{m,\text{dec}}}_{\text{decision delay}} + \underbrace{\mathcal{D}_{i,j,k}^{q,\text{fwd}} + \mathcal{D}_{i,j,k}^{\text{pro}} + \mathcal{D}_{i,j,k}^{\text{trans}}}_{\text{forwarding delay}}. \quad (10)$$

Assume that the collection of involved LEO satellites for routing packet $\mathcal{P}_k$ in the path from its source node to its destination node are denoted as $\mathcal{S}'_k$, and $\mathcal{S}'_k \subseteq \mathcal{S}$. If $\mathcal{P}_k$ is not dropped along this path, the total end-to-end routing delay for $\mathcal{P}_k$ can be presented as

$$\mathcal{D}_k = \sum_{Sat_i \in \mathcal{S}'_k} \mathcal{D}_{i,k}. \quad (11)$$

Notably, we focus on intersatellite routing in space in this study; therefore, the routing delay model described above does not apply to ground-to-satellite communication links.

V. FLOW-CENTRIC DRL APPROACH

In this section, we formulate the routing problem as a POMDP, thus eliminating the requirement of obtaining global network states of large-scale LSBN. We present a fully distributed MADRL algorithm that incorporates *double DQNs* (DDQNs) for effective policy learning in high-dynamic environments. The principal contribution lies in the elaboration of an innovative flow-centric DRL approach, which introduces a novel definition of traffic flow. It enables each satellite to employ DNN model inferences for routing traffic flows rather than individual packets. This approach can substantially mitigate the negative impact of DNN inference time, ultimately achieving high-throughput routing in LSBNs.

A. Partially Observable Markov Decision Process

Each LEO satellite is assumed to be an independent DRL agent making routing decisions for arriving packets based on its local observations. Since acquiring global network state in real time is practically impossible in large-scale LSBNs, this fully distributed routing problem on each satellite can be modeled as a POMDP, defined by a six-tuple

$$\mathcal{P} = (\mathcal{S}, \mathcal{A}, \mathcal{T}, \mathcal{R}, \mathcal{O}, \gamma). \quad (12)$$

Here, $\mathcal{S} = \{s_1, s_2, \dots, s_n\}$ denotes the state space, $\mathcal{A} = \{a_1, a_2, \dots, a_n\}$ indicates the action space. The matrix $\mathcal{T} = T(\mathcal{S}_{t+1} = s' | \mathcal{S}_t = s, \mathcal{A}_t = a)$ represents the conditional transition function. The reward function is denoted by $\mathcal{R} = \mathcal{S} \times \mathcal{A}$, and $\mathcal{O} = \{o_1, o_2, \dots, o_n\}$ signifies the local observation space of the DRL agent. The reward discount factor is denoted as $\gamma \in [0, 1]$. In greater detail, the specifics of observation, action, and reward in this study are defined as follows:

1) *Observation Space*: The observation vector of satellite Sat_i comprises several components, and is denoted as $\mathcal{O}_i = \{\Phi_{i,k}, \mathcal{B}_i, \Theta_i, \Omega_i\}$. Here, $\Phi_{i,k} = \{Dis_{i,k}^{\text{upper}}, Dis_{i,k}^{\text{lower}}, Dis_{i,k}^{\text{left}}, Dis_{i,k}^{\text{right}}\}$ denotes the spatial distances from the four neighboring satellites of Sat_i to the destination satellite of the current packet $\mathcal{P}_k$, which can be calculated using (8). It is feasible to estimate the spatial positions of the neighboring and the destination satellites using the existing SGP4 model [62]. We use $\mathcal{B}_i = \{b_i^{\text{upper}}, b_i^{\text{lower}}, b_i^{\text{left}}, b_i^{\text{right}}\}$ to indicate the available network bandwidths of the four ISLs connecting satellite Sat_i and its neighbors. In addition, the current traffic loads of the four forwarding queues on satellite Sat_i are $\Theta_i = \{\theta_i^{\text{upper}}, \theta_i^{\text{lower}}, \theta_i^{\text{left}}, \theta_i^{\text{right}}\}$, and the loads of the decision queues on Sat_i 's four neighbors are denoted as $\Omega_i = \{\omega_i^{\text{upper}}, \omega_i^{\text{lower}}, \omega_i^{\text{left}}, \omega_i^{\text{right}}\}$. As these factors span different value ranges, normalization is applied as expressed in the following equation:

$$x_{\text{norm}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}}. \quad (13)$$

2) *Action Space*: When routing packet $\mathcal{P}_k$, Sat_i can conduct one of the actions in $\mathcal{A}_i = \{a_i^{\text{upper}}, a_i^{\text{lower}}, a_i^{\text{left}}, a_i^{\text{right}}\}$, where $a_i^{\text{upper}}, a_i^{\text{lower}}, a_i^{\text{left}}$, and a_i^{right} represent the probabilities of selecting one of its four neighboring satellites as the next-hop.

3) *Reward Function*: To ensure that each agent, i.e., LEO satellite, to learn optimal routing decision, the reward function for Sat_i routing packet $\mathcal{P}_k$ at time t is defined as

$$r_i(t) = \begin{cases} -\psi, & \mathcal{P}_k \text{ is dropped} \\ -\kappa_1 Dis_{j,k} - \kappa_2 \mathcal{D}_{i,j,k}^{\text{fwd}} - \kappa_3 \mathcal{D}_{j,k}^{\text{dec}}, & \text{otherwise} \end{cases} \quad (14)$$

where $Dis_{j,k}$ indicates the normalized spatial distance between the next-hop satellite Sat_j and the destination satellite. The notation $\mathcal{D}_{i,j,k}^{\text{fwd}}$ denotes the normalized forwarding delay, and ψ is a penalty value to the agent in case of packet loss. It is noted that the normalized decision delay $\mathcal{D}_{j,k}^{\text{dec}}$ of routing packet $\mathcal{P}_k$ on Sat_j is further taken into account, which is overlooked in the previous studies. Additionally, κ_1, κ_2 and κ_3 are weights to balance the above factors. In order to account for the long-term impacts of actions, the cumulative discounted reward is calculated by

$$R_i(t) = \sum_{n=0}^{\infty} \gamma^n r_i(t+n) \quad (15)$$

where the discount factor γ is utilized to strike a balance between immediate rewards and future rewards.

B. Multiagent Deep Reinforcement Learning Algorithm

To maximize the gross cumulative discounted rewards and thereby to optimize the routing policy learned by each LEO satellite, a MADRL algorithm is derived. As shown in Fig. 4, The MADRL algorithm utilizes the DDQN approach [63] and is implemented in a fully distributed way. Each LEO satellite Sat_i utilizes two DNNs, i.e., an estimation Q -network $Q_i(o_i, a_i; \mu_i)$ and a target Q -network $Q'_i(o_i, a_i; \mu'_i)$, parameterized by μ_i and μ'_i , respectively. At each decision time t , satellite Sat_i employs local observation $o_i(t)$ to select an

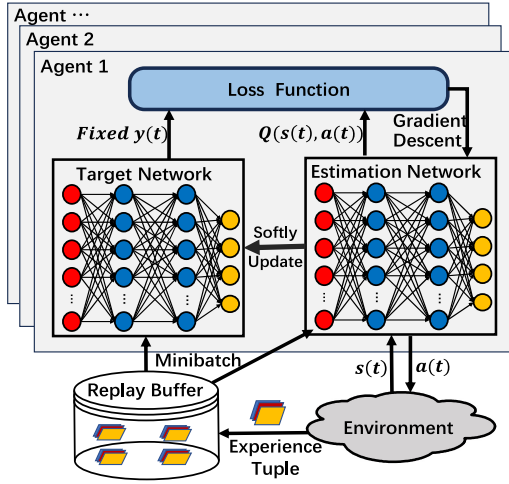


Fig. 4. Framework of proposed MADRL algorithm.

action $a_i(t)$ from the action space A_i based on the ε - greedy policy [64]

$$a_i(t) = \begin{cases} \text{random action,} & \text{propability} = \varepsilon \\ \text{argmax}_a Q_i(o_i(t), a_i(t); \mu_i), & \text{probability} = 1 - \varepsilon. \end{cases} \quad (16)$$

The current state is then changed to the next state $o_i(t+1)$, while satellite Sat_i will receive the reward $r_i(t)$ from environmental feedback. An experience tuple $\{o_i(t), o_i(t+1), a_i(t), r_i(t)\}$ will be recorded into the replay buffer RB , which is adopted to break the correlations of the training data, and thereby to optimize the training process. The agents are trained by randomly sampling a minibatch from RB and then updating the parameter values of the Q -networks. Particularly, the target Q -network is utilized to calculate a fixed target Q -value $y_i(t)$ for each state-action pair $(o_i(t), a_i(t))$ during multiple updates in each iteration

$$y_i(t) = r_i(t) + \gamma \max_{a_i(t+1)} Q'_i(o_i(t+1), a_i(t+1); \mu'_i) \quad (17)$$

where $r_i(t)$ is the immediate reward of $(o_i(t), a_i(t))$. The loss function $\text{Loss}_i(t)$ is defined as follows:

$$\text{Loss}_i(t) = (y_i(t) - Q_i(o_i(t), a_i(t); \mu_i))^2. \quad (18)$$

The estimation Q -network is trained to minimize the loss following the gradient descent algorithm:

$$\mu_i = \mu_i + \alpha \nabla_{\mu_i} \text{Loss}_i(t) \quad (19)$$

where α is the learning rate (LR). After each iteration, rather than directly copying the parameters of the estimation network, the target Q -network will be softly updated

$$\mu'_i = \tau \mu_i + (1 - \tau) \mu'_i \quad (20)$$

where $\tau \in (0, 1]$ is a hyperparameter that serves to limit the pace of change in the target Q -network, significantly enhancing the stability of the learning process [65]. Gradually, the Q -networks can estimate the agent's routing decisions more accurately.

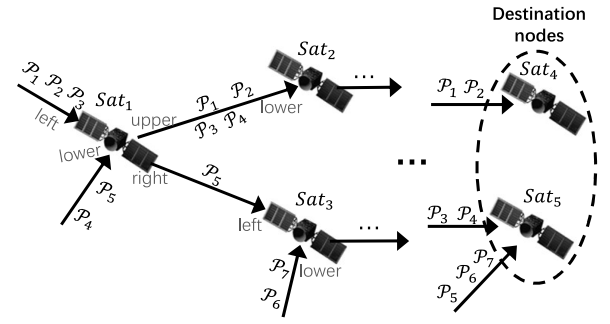


Fig. 5. Example scenario to elucidate the novel definition of traffic flow.

C. Flow-Centric DRL Approach

In order to counteract the detrimental effect of DNN model inference time on routing throughput within LSBNs, this study introduces a novel DRL approach leveraging the flow-centric strategy. This approach focuses on routing traffic at the granularity of traffic flows rather than individual packets. The concept of a traffic flow is traditionally understood as a sequential arrangement of data packets with the same origin and destination nodes. However, LSBN serves a global user base, with packets originating from diverse users on the ground and destined for target users. The multitude of source-destination user pairs lead to an explosion in the flow count, causing unacceptable overhead for on-board DNN inferences. Thus, specific tailoring of the flow definition is necessary to better suit the system.

This study centers on intersatellite routing and does not encompass satellite-to-ground transmissions. Consequently, the satellites situated in space, which serve as the access points for source and destination ground users, are designated as the source and destination nodes in the context of defining traffic flows. Additionally, given that the formulated POMDP necessitates independent routing decisions by each LEO satellite based on local observations, precise knowledge of source locations holds limited significance. Thus, each LEO satellite treats its respective inbound port—where packets are received—as the source for a packet. With these designs, the number of established flows on each satellite is significantly reduced. Consequently, the following definition of the term flow is established within the context of this study:

Definition 1: On each LEO satellite router, a traffic flow pertains to a consecutive sequence of traffic packets, which are received from the same networking port and destined for the same LEO satellite node.

In further elucidation of the concept, we employ illustrative scenarios, as depicted in Fig. 5. A sequence of data packets— $P_1, P_2, P_3, P_4, P_5, P_6, P_7$, and so forth—progress through relays on various LEO satellites, including Sat_1 , Sat_2 , and Sat_3 , among others, culminating in delivery to destination LEO satellites, namely, Sat_4 and Sat_5 . The exact definitions of these flows, aligned with Definition 1, are meticulously presented in Table I. It is pertinent to note that, within the fully distributed routing framework, differing LEO satellites hold distinct interpretations of traffic flows. For instance, from the viewpoint of satellite Sat_1 , P_3 belongs to flow $^2_{Sat_1}$, while P_4

TABLE I
SPECIFIC DEFINITION OF FLOW IN EXAMPLE SCENARIO

	Flow ID	Inbound Port	Destination Node	Included Packets
Sat_1	$flow_{Sat_1}^1$	Left-link	Sat_4	$\mathcal{P}_1, \mathcal{P}_2, \dots$
	$flow_{Sat_1}^2$	Left-link	Sat_5	$\mathcal{P}_3, \dots$
	$flow_{Sat_1}^3$	Lower-link	Sat_5	$\mathcal{P}_4, \mathcal{P}_5, \dots$
Sat_2	$flow_{Sat_2}^1$	Lower-link	Sat_4	$\mathcal{P}_1, \mathcal{P}_2, \dots$
	$flow_{Sat_2}^2$	Lower-link	Sat_5	$\mathcal{P}_3, \mathcal{P}_4, \dots$
Sat_3	$flow_{Sat_3}^1$	Left-link	Sat_5	$\mathcal{P}_5, \dots$
	$flow_{Sat_3}^2$	Lower-link	Sat_5	$\mathcal{P}_6, \mathcal{P}_7, \dots$

is associated to $flow_{Sat_1}^3$. In contrast, satellite Sat_2 categorizes both $\mathcal{P}_3$ and $\mathcal{P}_4$ under $flow_{Sat_2}^2$.

In the literature, the packet-centric DRL approach is already widely used to implement traffic routing in LSBNs. A flow-centric DRL approach is proposed in this article. Considering that all the packets in a traffic flow have the same destination address, it is specified that each LEO satellite independently maintains a *flow routing table* for all the traffic flows passing through it. Similar to the conventional packet-centric DRL approach, the flow-centric DRL approach requires an LEO satellite to utilize DNN model to make routing decision for the first packet in a traffic flow. The route information is then stored as a corresponding entry in the LEO satellite's flow routing table, and can be directly used by the subsequent packets in the same flow. This can effectively eliminate the needs of conducting DNN model inferences for the subsequent packets, and therefore can significantly reduce the cumulative time spent on DNN model inferences in LSBNs. As a consequence, the routing performance, such as end-to-end transmission delay, packet loss, particularly satellite, and network throughputs, are anticipated to experience substantial optimization, meeting the high-throughput demands of large-scale LSBNs. To summarize, the proposed flow-centric DRL approach for high-throughput routing in LSBNs is encapsulated in the algorithm described in Algorithm 1.

VI. ADAPTIVE FLOW ROUTING UPDATING SCHEME

A. Motivation

While the proposed flow-centric DRL approach focuses on routing traffic flows, it's essential to recognize that satellites primarily operate by forwarding individual packets at a granular level, representing a continuous and sequential routing process. However, LSBN exhibits high dynamism and time-varying characteristics that can undermine the performance and reliability of the flow-centric DRL approach. Specifically, network state changes, such as link interruptions, coverage variations, or load fluctuations, can render previously established routing decisions for a traffic flow obsolete for the subsequent packets.

To illustrate this issue, we implement the proposed flow-centric DRL approach in a simulated *Iridium*-like network [59]. This network consists of 21 000 terrestrial users,

Algorithm 1 Flow-Centric Multiagent DRL-Based Routing Approach in LSBN

Initialization: Initialize estimation Q -networks, target Q -networks and the replay buffer RB ; initialize the Flow Routing Table on each satellite

```

1: for  $step \leftarrow 1$  to  $\max - \text{train} - \text{steps}$  do
2:    $Sat_i$  receives packet  $\mathcal{P}_k$ 
3:    $Sat_i$  makes routing decision  $a_i(t)$  based on local observation  $o_i(t)$  and the  $\varepsilon - \text{greedy}$  policy (16)
4:    $Sat_i$  gets an immediate reward  $r_i(t)$ 
5:   Store the tuple  $\{o_i(t), o_i(t+1), a_i(t), r_i(t)\}$  to  $RB$ 
6:    $step \leftarrow step + 1$ 
7:   Randomly sample a minibatch from  $RB$ 
8:   Update estimation  $Q$ -networks by minimizing loss (18)
9:   Soft update target  $Q$ -networks by (20)
10: end for
11:  $terminal \leftarrow \text{False}$ 
12: while  $\text{not } terminal$  do
13:    $Sat_i$  receives packet  $\mathcal{P}_k$  and identify the flow  $flow_{Sat_i}^v$  to which  $\mathcal{P}_k$  belongs, based on Definition 1
14:   if  $\mathcal{P}_k$  is the first packet of  $flow_{Sat_i}^v$  then
15:      $Sat_i$  employs the DNN to make routing decision
16:     Store the route for  $flow_{Sat_i}^v$  in Flow Routing Table
17:   else
18:     Use the route stored in Flow Routing Table
19:   end if
20:   if all packets are routed then  $terminal \leftarrow \text{True}$  end if
21: end while

```

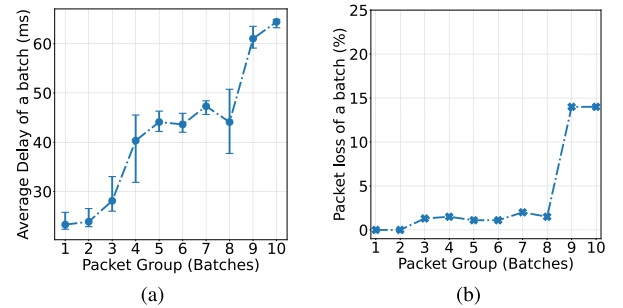


Fig. 6. Average end-to-end delay and packet loss rate for each batch within the chosen traffic flow. (a) Average delay of packets within each group. (b) Packet loss within each group.

each continuously sending data at an average rate of 10 pps. Each ISL has a maximum bandwidth of 1.2 Gb/s. Decision and forwarding queues have upper bounds of 400 and 200, respectively. Parameters regarding the structure and scale of the DNN are kept consistent with those outlined in Section III for experimental consistency.

We select a specific traffic flow and divide its initial 1000 packets into 10 groups, each containing 100 packets as a *batch*. As shown in Fig. 6(a), the routing decision made by the proposed flow-centric DRL approach performs excellently for the initial batches, maintaining average delays consistently below 30 ms. However, its effectiveness gradually diminishes, particularly from the 9th batch onward, where the average delay exceeds 60 ms. Similarly, Fig. 6(b) illustrates the packet

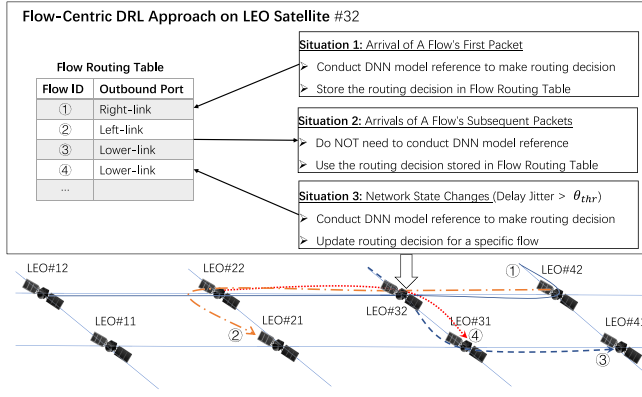


Fig. 7. Principle of adaptive flow routing updating scheme in LSBN.

loss ratios, with a noticeable deterioration occurring suddenly in the last two batches. Coupled with the challenge of adapting to network dynamics, the flow-centric DRL approach encounters hurdles in maintaining high performance in LSBN. Therefore, it is imperative for each LEO satellite to continuously evaluate whether the previously established routes stored in its flow routing table remain optimal for the ongoing flow transmissions in refreshed network states. If not, LEO satellites should promptly update the stored routes based on current network state information.

B. Delay Jitter-Based Updating Scheme

The core of flow routing updates lies in enabling satellites to identify the appropriate moments for route updates. However, in highly dynamic and large-scale LSBNs, numerous factors significantly impact the performance of the flow-centric DRL approach, making it exceedingly complex and challenging to identify these influences accurately. Consequently, collecting a substantial volume of real-time network state change information for each LEO satellite is a nontrivial task. This complexity makes it unfeasible to rely solely on network models for flow routing updates in LSBN.

To address this challenge, we introduce an adaptive flow routing updating scheme based on the *delay jitter* metric. This scheme is entirely model-free and does not require complex network state monitoring mechanisms. As depicted in Fig. 7, Sat_i only needs to measure and record the packet forwarding delay $\mathcal{D}_i^{v,k}$ after successfully forwarding a packet $\mathcal{P}_k$ in flow $flow_{Sat_i}^v$. Subsequently, it can derive the delay jitter $\Delta\mathcal{D}_i^{v,k}$ by calculating the difference between the forwarding delays of the two consecutive data packets, $\mathcal{P}_k$ and $\mathcal{P}_{k-1}$, in flow $flow_{Sat_i}^v$.

$$\Delta\mathcal{D}_i^{v,k} = \left| \mathcal{D}_i^{v,k} - \mathcal{D}_i^{v,k-1} \right|. \quad (21)$$

By comparing the delay jitter $\Delta\mathcal{D}_i^{v,k}$ with a predefined threshold θ_{thr} , Sat_i can assess the current suitability of the route stored in its flow routing table for flow $flow_{Sat_i}^v$. If the delay jitter exceeds the threshold, when $\mathcal{P}_{k+1}$ arrives, Sat_i will trigger the DNN model inference process to calculate a new

Algorithm 2 Flow-Centric DRL Approach With Adaptive Flow Routing Updating Scheme in LSBN

Initialization: Initialize the Flow Routing Table and well-trained DNN model(s) on each LEO satellite; Initialize delay jitter threshold θ_{thr}

```

1: terminal  $\leftarrow$  False
2: while not terminal do
3:    $Sat_i$  receives packet  $\mathcal{P}_k$  belonging to flow  $flow_{Sat_i}^v$ 
4:   if  $\mathcal{P}_k$  is the first packet of  $flow_{Sat_i}^v$  then
5:      $Sat_i$  employs the DNN to make routing decision
6:     Store the route for  $flow_{Sat_i}^v$  in Flow Routing Table
7:   else
8:     if  $\Delta\mathcal{D}_i^{v,k} > \theta_{thr}$  then
9:        $Sat_i$  employs the DNN to make routing decision
10:      Update route for  $flow_{Sat_i}^v$  in Flow Routing Table
11:    else
12:      Use the route stored in Flow Routing Table
13:    end if
14:  end if
15:  Perceive and record the routing delay  $\mathcal{D}_i^{v,k}$  of  $\mathcal{P}_k$ 
16:  Calculate the delay jitter  $\Delta\mathcal{D}_i^{v,k}$  of  $flow_{Sat_i}^v$  by (21)
17:  if all packets are routed then terminal  $\leftarrow$  True end if
18: end while

```

route for $flow_{Sat_i}^v$ using the following equation:

$$route_i^{v,k+1} = \begin{cases} \operatorname{argmax}_a Q_i(o_i(t), a_i(t); \mu_i), & \Delta\mathcal{D}_i^{v,k} > \theta_{thr} \\ route_i^{v,k}, & \text{otherwise} \end{cases} \quad (22)$$

where $Q_i(o_i(t), a_i(t); \mu_i)$ is the well-trained Q -network on Sat_i to make routing decisions. The new route and the outdated one for $flow_{Sat_i}^v$ are represented by $route_i^{v,k+1}$ and $route_i^{v,k}$, respectively. The new route is then used to forward the subsequent packets $\{\mathcal{P}_{k+1}, \mathcal{P}_{k+2}, \dots\}$ in $flow_{Sat_i}^v$. In summary, the flow-centric DRL approach with the adaptive updating scheme for LSBN is presented in Algorithm 2.

VII. PERFORMANCE EVALUATION

A. Performance Metrics

We utilize N and N_{loss} to indicate the total number of packets and the number of dropped packets, respectively. During the simulation duration T , the volume of data forwarded by Sat_i is denoted as $\mathcal{V}_i$, and the overall data volume passing through the entire network is represented by $\mathcal{V}$. The following four performance metrics of the derived flow-centric DRL approach are evaluated in this article.

- 1) *End-to-End Delay* ($\mathcal{D}_{e2e}$): The average delay for delivering a packet from a source satellite to a destination satellite, which can be calculated by

$$\mathcal{D}_{e2e} = \frac{\sum_{k=1}^{N-N_{loss}} \mathcal{D}_k}{N - N_{loss}}.$$

- 2) *Packet Loss* ($\mathcal{R}_{loss}$): The ratio of the dropped packets among all the packets transmitted in the LSBN, which can be expressed as

$$\mathcal{R}_{loss} = \frac{N_{loss}}{N}.$$

- 3) *Satellite Throughput* (Tp_{sat}): The maximum volume of traffic that a single satellite can route within a time unit

$$Tp_{sat} = \max_{Sat_i} \frac{V_i}{T}.$$

- 4) *Network Throughput* (Tp_{net}): The volume of traffic that can be transmitted through the entire LSBN within a time unit, indicated by

$$Tp_{net} = \frac{V}{T}.$$

B. Experiment Setup

We developed a discrete-event-driven simulator from scratch using *STK* and *Python*. This simulator can replicate ground user distribution, user behavior, traffic generation and transmission, queuing events, as well as dynamic topology in LSBN. *PyTorch* is utilized to implement the MADRL algorithm, including DNN model training and inference. Furthermore, to fully showcase robustness of the proposed flow-centric DRL approach, we conducted extensive experiments encompassing two representative scenarios, with different satellite constellation configurations and user counts.

- 1) *Iridium-Like LSBN*: *Iridium*, as a representative LEO satellite communication system, has been referenced in numerous LSBN-related academic research [5], [38], [66]. In line with these studies, we draw on the configuration of the Iridium constellation to design an Iridium-like LSBN scenario. There are 66 LEO satellites at an altitude of 780 km, distributed across 6 orbits, forming a 6×11 *Walker* constellation. The orbital inclination is set at 86.4 degrees, with an eccentricity of 0. The user count in this scenario spans from 15 000 to 27 000.
- 2) *Large-Scale LSBN*: In the next generation of space Internet, attention has been directed toward mega constellations [6]. To keep up with these developments, we simulate a large-scale LSBN scenario, which substantially augments the number of satellites, comprising a more extensive 12×30 *Walker* constellation with a lower orbital altitude of 500 km. And the number of users is scaled up, ranging from 30 000 to 50 000. Other settings remain consistent with those of the Iridium-like LSBN.

The network traffic in the above scenarios is generated based on the user model established in Section IV-B. We leverage the Gridded Population of the World data set released by NASA [61] to estimate user density within different regions. And the request frequency of each user follows a negative exponential distribution with an expectation of 10 pps. The destination nodes for these packets are chosen completely at random. The capability of each satellite to conduct DNN inference in parallel is set to be 8, and the network bandwidth of each ISL is 1.2 Gb/s. The upper bounds of decision and forwarding queue on each satellite are set at 400 and 200, respectively. The simulation duration is 60 min, during which each LEO satellite can complete more than half an orbit around the Earth, encompassing a significant amount of topology and

TABLE II
HYPERPARAMETERS AND CONFIGURATIONS OF MADRL

Hyper Param.	Values	Hyper Param.	Values
Num. of episodes	10	Discount factor (γ)	0.99
Num. of network layers	3	Action space	discrete
Learning rate	0.0005	Batch size	128
Num. of neurons per layer	256	Optimizer	Adam
Replay buffer size	500,000	Training frequency	10

state dynamics. Table II provides the hyperparameters and configurations for our proposed approach. For the given size of DNN, the inference time is configured by measuring on a computer with Intel Core i7-11800H and 16 GB of RAM.

C. Baseline Algorithms

Aiming to comparatively evaluating the performance of the derived flow-centric DRL approach, this article adopts four representative baseline algorithms.

- 1) *Open Shortest Path First (OSPF) Routing Algorithm*: OSPF periodically computes the shortest paths from the current router to various destination nodes, based on the distances measured with unit weights (hop count), and then stores them in the routing table. When a packet arrives at the router, routing decisions are made by referencing the routing table, and the decision delay can be considered negligible.
- 2) *Weight Optimization OSPF (HeurOSPF)*: HeurOSPF [67] is an enhancement of OSPF that equips OSPF with abilities to avoid congestion on ISLs. HeurOSPF utilizes a local search heuristic to compute appropriate link weight settings that achieve promising load balancing performance with equal-cost multipath routing (ECMP).
- 3) *ELB*: routing algorithm. ELB [38] was designed to mitigate load imbalance in satellite networks. Similar to OSPF, the routing table is computed in a manner that allows the routing decision delay of the ELB algorithm to be disregarded.
- 4) *Packet-Centric DRL Approach*: In line with the concepts presented in prior research [31], [32], an LEO satellite employs a separate DNN model inference for routing each traffic packet, which introduces nonnegligible cumulative decision delay.

D. Results Discussion

We conducted comparative experiments for two scenarios: 1) Iridium-like and 2) large-scale LSBNs, each with varying numbers of ground users. In addition to evaluating the four baseline algorithms, this article focuses on assessing the performance of the proposed flow-centric DRL approach under three different delay jitter thresholds: 1) θ_{thr} set as 0.001; 2) 0.005; and 3) 0.008, respectively. The convergence results in two scenarios, with varying values of hyperparameters, are shown in Figs. 8 and 9. The performance evaluation results in two scenarios are depicted in Figs. 10 and 11, respectively. It is noted that, in both scenarios, the proposed flow-centric DRL approach consistently demonstrates superior performance compared to all the baseline algorithms across various metrics, including end-to-end delay, packet loss, and

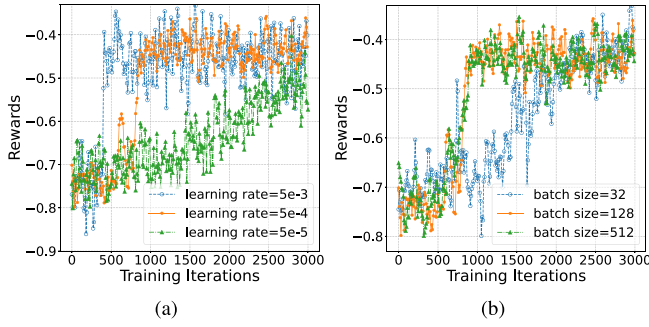


Fig. 8. Convergence diagram of the MADRL model in the Iridium-like LSBN environment under various LR and batch size. (a) Batch size is 128. (b) LR is fixed at 0.0005.

network throughput. In terms of satellite throughput, the flow-centric DRL approach achieves significant optimizations when compared to the packet-centric approach, bridging its performance closer to that of OSPF, HeurOSPF and ELB.

1) *Convergence and Hyperparameter Analysis:* The assessment of the convergence in DRL primarily depends on observing the rewards obtained from the actions taken by the agents. Fig. 8 shows the reward convergence of the MADRL model during the training process. With a LR set to 0.0005 and a batch size of 128, after 1000 training iterations, the reward of agents converges to approximately -0.4 . Fig. 8(a) illustrates the impact of the LR on convergence performance. It can be seen that a small LR leads to slow training, with the reward converging to around -0.5 after 2700 iterations, failing to learn the optimal routing policy. On the contrary, a large LR results in fast convergence, but there are significant oscillations, leading to performance instability. Additionally, as depicted in Fig. 8(b), slow convergence and substantial oscillations occur when the batch size is decreased to 32. As the batch size increases, the convergence performance improves. Observations suggest that the convergence performance is similar when the batch size is set to 128 and 512, respectively. Considering that a greater batch size causes more time consumption per training iteration, setting it at 128 is a better choice in this study. Besides, we recorded the convergence process in the large-scale LSBN, as illustrated in Fig. 9. It is evident that, with a significant expansion in network scale, the convergence speed of our MADRL model only slightly decelerates and remains acceptable. This demonstrates the scalability of our proposed distributed approach.

2) *Performance Factors Analysis:* The reason behind the superior network throughput performance of the flow-centric DRL approach over the OSPF and ELB algorithms lies in its consideration of not only the spatial distance from the next-hop satellite to the destination satellite but also the real-time traffic states on the corresponding ISLs. The flow-centric DRL approach strives to achieve a better balance between propagation and queuing delays by harnessing the power of DNN models. It also exhibits better adaptability to dynamic LSBN environments, allowing for improved performance and efficiency. Owing to the effective link congestion mitigation, HeurOSPF outperforms other baselines and achieves comparable performance to our approach in terms of packet loss rate

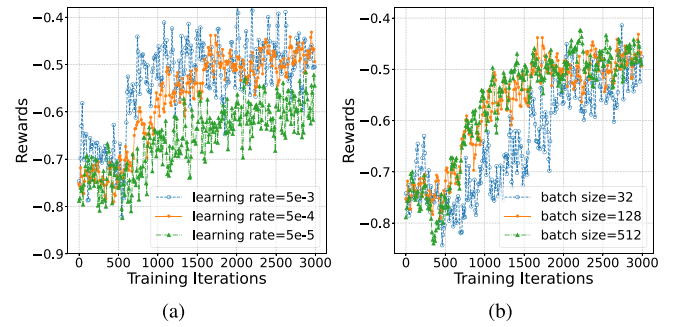


Fig. 9. Convergence diagram of the MADRL model in the large-scale LSBN environment under various LR and batch size. (a) Batch size is 128. (b) LR is fixed at 0.0005.

and network throughput. However, due to the restriction of the equal split rule, HeurOSPF considers paths of different lengths as equal, resulting in increased propagation distances and inefficient utilization of bandwidth. This limitation makes it inferior to our approach in terms of end-to-end delay. Remarkably, HeurOSPF is a heuristic algorithm designed for terrestrial fixed networks, which relies on the pre-established traffic matrix (TM) and substantial iterative local searches to obtain optimized weights. In the highly dynamic and large-scale LSBN, real-time measurements of ever-changing global TMs are challenging, and the time consumption of iterative optimization makes it impractical for deployments in LSBN. It is worth noting the packet-centric DRL approach exhibits the poorest performance among the four baseline algorithms. This is due to the fact that every packet necessitates a separate DNN model inference, and the cumulative inference delay significantly degrades routing performance. In contrast, the flow-centric DRL approach implements a traffic flow-based routing mechanism, allowing packets within the same flow to share the same route information stored in the flow routing table. This substantially reduces the need for DNN model inferences and effectively mitigates the issues associated with DRL-based routing algorithms, as described in Section III.

3) *Performance Variations Between Different Scenarios:* The evaluated performance differs between the Iridium-like and the large-scale LSBN scenarios due to several factors. First, the large-scale LSBN encompasses much more satellites than the Iridium-like LSBN. Consequently, packets in the large-scale LSBN traverse a greater number of hops in their routing paths, which leads to larger cumulative decision and forwarding delays. Therefore, as depicted in Figs. 10(a) and 11(a), the End-to-End Delay in the large-scale LSBN is higher than that in the Iridium-like LSBN. On the other hand, when comparing Fig. 10(d) with Fig. 11(d), it becomes evident that the large-scale LSBN exhibits a higher potential for network throughput compared to the Iridium-like LSBN. This is primarily attributed to the fact that a large-scale network typically possesses greater total bandwidth resources and have a broader reach in terms of accessing ground users.

4) *Analyzing Adaptive Flow Routing Updating Scheme:* Setting an appropriate value for the delay jitter threshold θ_{thr} to match the distinct characteristics of various scenarios

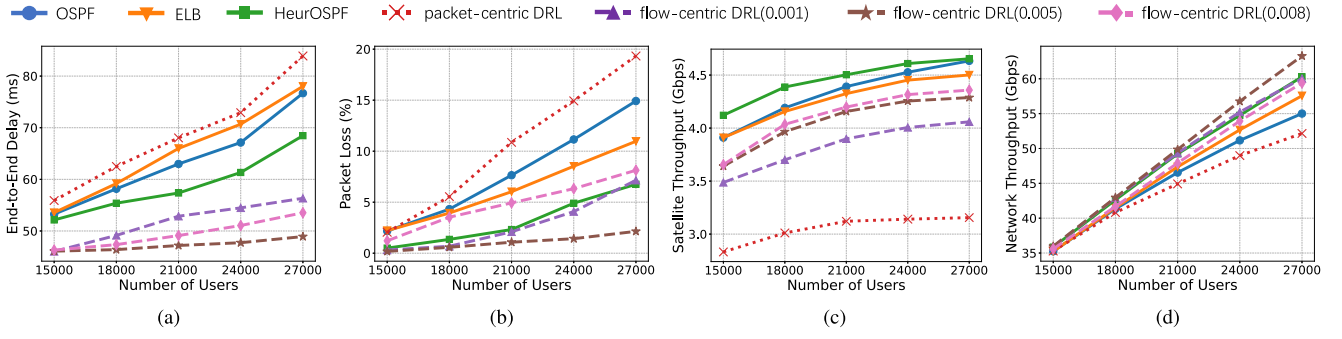


Fig. 10. Comparative experimental results against different number of ground users in Iridium-like LSBN. (a)–(d) Experimental results for end-to-end delay, packet loss, satellite throughput, and network throughput.

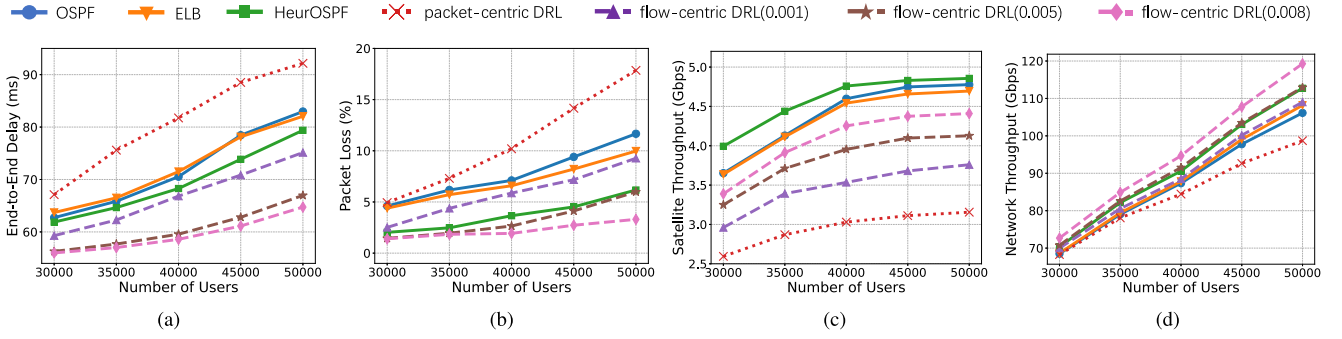


Fig. 11. Comparative experimental results against different number of ground users in large-scale LSBN. (a)–(d) Experimental results for end-to-end delay, packet loss, satellite throughput, and network throughput.

TABLE III
EVALUATION RESULTS OF DECISION DELAY IN TWO SCENARIOS

Metrics	Average end-to-end decision delay		Proportion of decision delay	
	Iridium-like LSBN (15000 / 27000)	Large-scale LSBN (30000 / 50000)	Iridium-like LSBN (15000 / 27000)	Large-scale LSBN (30000 / 50000)
Packet-centric DRL	8.5 ms / 42.9 ms	17.9 ms / 51.9 ms	15.1 % / 51.1 %	26.7 % / 56.3 %
Flow-centric DRL (0.001)	2.2 ms / 6.1 ms	9.8 ms / 26.4 ms	4.7 % / 10.8 %	16.5 % / 35.2 %
Flow-centric DRL (0.005)	1.2 ms / 1.5 ms	3.7 ms / 12.5 ms	2.6 % / 3.0 %	6.6 % / 18.7 %
Flow-centric DRL (0.008)	0.9 ms / 1.1 ms	2.1 ms / 3.0 ms	1.9 % / 2.1 %	3.8 % / 4.6 %
Other Algorithms	0.0 ms / 0.0 ms	0.0 ms / 0.0 ms	0.0 % / 0.0 %	0.0 % / 0.0 %

is crucial for the adaptive flow routing updating scheme. We investigate the performance of the proposed flow-centric DRL approach by setting three different values for θ_{thr} : 1) 0.001; 2) 0.005; and 3) 0.008. As depicted in Fig. 10, for the Iridium-like LSBN scenario, the flow-centric DRL (0.005) outperforms both flow-centric DRL (0.001) and flow-centric DRL (0.008). The intuitive reason is that when θ_{thr} is too large, the updating of the flow routing table may lag behind the real-time network changes. Conversely, a too-small value of θ_{thr} can result in an increased demand for DNN model inferences, thereby hampering the performance of the proposed approach. However, different results are observed in the large-scale LSBN scenario. As shown in Fig. 11, the performance of flow-centric DRL (0.008) slightly outperforms that of flow-centric DRL (0.005). Our analysis suggests that in the large-scale LSBN, the traffic flows defined by satellite are more diverse and abundant in quantity, leading to a larger number of routes for these flows. Thus, the adjustment of further increasing θ_{thr} to maintain a sufficiently optimized DNN inference frequency becomes imperative when applying the

proposed flow-centric DRL approach in the large-scale LSBN scenario.

5) *Average Decision Delay Optimization*: We further investigate the extent to which the proposed flow-centric DRL approach can optimize the *average decision delay* (ADD) for each packet. The evaluation results in the two scenarios are outlined in Table III. The ADD of OSPF and ELB can be ignored because looking up the route table is typically instantaneous. When dealing with a larger number of users, i.e., 27 000 users in the Iridium-like LSBN and 50 000 users in the large-scale LSBN, the ADD incurred by the packet-centric DRL approach reaches 42.9 ms and 51.9 ms, accounting for more than half of the total end-to-end delay. In contrast, the flow-centric DRL approach (with $\theta_{thr} = 0.008$) can significantly reduce the ADD to around 1 ms and 3 ms in the two scenarios. Additionally, the proportion of the decision delay when using the flow-centric DRL approach is dramatically reduced as well, accounting for only around 2%

of the total end-to-end delay in the Iridium-like LSBN and 4.6% in the large-scale LSBN.

6) *Satellite and Network Throughput Optimization*: Through a reduction in DNN inferences and decision delay, the flow-centric DRL approach can significantly alleviate the throughput constraints described in Section III. Regarding satellite throughput, as depicted in Figs. 10(c) and 11(c), the packet-centric DRL approach faces limitations stemming from inference capacity and delay, which pose challenges in enabling satellites to achieve sufficient throughput. At the same time, the flow-centric DRL approach shows its effectiveness in optimizing satellite throughput, approaching levels comparable to OSPF and ELB algorithms. As θ_{thr} decreases, the enhancement in satellite throughput becomes less pronounced. The throughput of individual satellites may not comprehensively reflect overall performance. As depicted in Figs. 10(d) and 11(d), the flow-centric DRL approach enables the entire LSBN to attain approximately 64 Gb/s and 118 Gb/s throughputs in the two scenarios, surpassing all the baselines. This effectively showcases its superior performance in empowering high-throughput routing in LSBN.

VIII. CONCLUSION

Existing research efforts on utilizing DRL for packet routing in LSBN usually neglect the DNN model inference time for making packet routing decisions. This article has diligently addressed this oversight and illustrated its detrimental impact on routing performance through extensive simulations. The primary contribution of this article lies in elaborating on a decentralized flow-centric DRL approach tailored for LSBN. Unlike traditional DRL approaches that focus on individual packet routing, our approach prioritizes routing for traffic flows, leading to a substantial reduction in the quantity of DNN inferences. Notably, the proposed flow-centric DRL operates within a fully distributed architecture where each satellite independently classifies local flows and decides next-hop routes for them, without relying on any central controllers. Additionally, we further introduce an adaptive and model-free flow routing updating scheme, enhancing the adaptability of the flow-centric approach in high-dynamic LSBN. Our comprehensive experimental results indicate: the flow-centric DRL approach outperforms the baseline algorithms across key performance metrics, including end-to-end delay, packet loss, network throughput, and ADD; it enhances the high-throughput routing capability within LSBN, thereby providing robust support for increased traffic transmission in future LSBN scenarios.

As a natural progression of our research, we intend to delve into the performance of the proposed approach in higher fidelity LSBNs characterized by more intricate constellation configurations. Moreover, we aspire to integrate the proposed flow-centric DRL approach with accelerators, including NPU, in the future, collaborating to achieve further enhancement of satellite throughput in large-scale LSBNs. Our approach are also intended to be applied in other large-scale scenarios with high-throughput demand, such as global video streaming and massive image production. These investigations promise to shed further light on the capabilities and potential

enhancements of our approach, ultimately contributing to the advancement of LSBN routing techniques.

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